

Machine-Learned Jet p_T Calibration from Low-Level Detector Information

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UTA Detector Foundation Model group

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Abstract

We present a systematic study of machine-learned jet transverse-momentum calibration from low-level detector information, using 620k truth-matched light jets from simulated $t\bar{t}$ events. A shared token encoder over calorimeter cells (with the detector neighbor graph recovered via exact cell matching between two ntuple formats), charged tracks, and reconstructed jet features is trained to regress the per-jet log p_T response with a heteroscedastic Gaussian objective. We compare all input combinations, graph versus set encodings of the calorimeter, and self-supervised masked-cell pretraining against from-scratch training. The best configuration (tracks + jet features + cells on the neighbor graph) approximately halves the jet energy resolution relative to the input reconstructed jets (11.9% versus 23.8% inclusive). Detector topology is a measurable inductive bias for cells-only models; combining all modalities requires $\mathcal{O}(10^5)$ labeled jets to pay off. A linear probe of the pretrained encoder matches full end-to-end training, demonstrating that the masked-cell pretext learns calibration-relevant representations; label-efficiency gains from pretrained initialization are small but consistent below 10% label fractions.

1 Introduction

This note is the first of three [1, 2] documenting the initial physics program of the UTA Detector Foundation Model (DFM) effort, which unifies two previously separate model lines—a calorimeter-cell graph network for topoclustering (M. A. Ghaznavi) and a track-based attention model for event-level b -tagging (U. Ghaznavi)—into a single shared backbone with task-specific heads (the `dfm` package). Hadronic jet calibration was selected as the first benchmark task following the program review [3]: it is a regression with unambiguous truth, a strong classical baseline (the ATLAS jet energy scale chain [4]), and it exercises every component of the merged architecture.

The target throughout is the per-jet log response $y = \ln(p_T^{\text{true}}/p_T^{\text{reco}})$; the model predicts (μ, σ) per jet, and the multiplicative correction e^μ defines the calibrated jet. The study is organized in gated tiers: dataset QA and no-training baselines (Tier 0), single-input models (Tier 1), input combinations and protocol cross-checks (Tier 2), the full-data matrix (Tier 3), and self-supervised pretraining (Tier 4).

2 Data samples and the cell-matching bridge

Two ntuple formats enter the study. The **5D ntuples** (17 files, 340k simulated $t\bar{t}$ events) provide tracks, jets, truth information, and a *thinned* calorimeter cell record; we measured the thinning empirically to be a per-cell threshold $E > 0.5 \text{ GeV}$ and $E > 2\sigma_{\text{noise}}$, positive energies only. The

Calo ntuples provide all 187,652 calorimeter cells per event in fixed hash order with the full detector neighbor list, plus topocluster associations.

The two formats contain different event samples, but their cell centres are bit-identical: an exact nearest-neighbour match in (x, y, z) within sampling layer (library `cell_matching.py`) identifies every 5D cell with its calorimeter hash position with zero mismatches in 146k jets tested, and a persistent Cell-ID lookup makes this free after warm-up. This bridge transfers the *detector neighbor graph* onto the thinned 5D cells—enabling the graph-encoding comparisons of this note—and connects the pretraining corpus (Calo format) to the calibration data (5D format).

The thinned cell record retains 75.0% of the energy of the cells in jets ($p_T > 20$ GeV; per-jet median 63%, 10th–90th percentile 32–85%, rising with jet energy), with strong layer dependence (EMB1/EMB3 $\sim 33\%$, Tile A/BC 81–86%). Cells-only results below are therefore lower bounds on full-cell performance. Large negative deposits in the forward calorimeter (FCAL pileup noise reaches tens of GeV per cell, and out-of-time pileup produces negative event sums) are handled with a cell-level $|E| < 100$ GeV cut rather than event vetoes; jet cones are central and unaffected.

3 Per-jet dataset

One sample per truth-matched light jet (`HadronConeExclTruthLabelID=0`): $p_T^{\text{true}} > 20$ GeV, recomputed $\Delta R(\text{reco}, \text{truth}) < 0.3$, $|\eta| < 2.5$, duplicate-truth-match removal. Isolation (no reco jet with $p_T > 15$ GeV within $\Delta R < 0.8$; no other truth match within $\Delta R < 1.0$) is stored as a flag: training uses isolated jets, and both populations are evaluated separately. Per jet we store: 6 jet features; all ghost-associated tracks with 18 features each (kinematics, impact-parameter significances computed from the track covariances, hit content); all cone ($\Delta R < 0.4$) cells with the 7-feature calorimeter schema and their matched neighbor subgraph; and metadata. Splits are by file: 13 train / 2 validation / 2 test files = 181k isolated training jets of 620k total. Training weights flatten the p_T^{true} spectrum over 20–250 GeV (capped at the 99th percentile).

4 Methodology

4.1 Model

All configurations are assemblies of the shared `dfm` backbone: per-modality input projections with type embeddings; local mixing per modality (SAGE-style convolution over the fixed calorimeter neighbor subgraph for cells; masked k -NN EdgeConv in $(\eta, \sin \phi, \cos \phi)$ for tracks); a shared mask-aware induced-set-attention (ISAB) stack, cross-modality where multiple inputs are present; masked-mean global pooling; and a head MLP emitting $(\mu, \log \sigma^2)$. Jet features, when enabled, join at the head. Models are ~ 1 M parameters (dim 128, 2 local + 2 global layers)—deliberately small, as the study compares information content, not capacity.

4.2 Training and metrics

Loss: Huber warm-up (5 epochs) then heteroscedastic Gaussian NLL; AdamW with cosine schedule; early stopping on validation NLL; 3 seeds per configuration. Metrics per $(p_T^{\text{true}}, |\eta|, \phi)$ bin on the corrected response $r_{\text{corr}} = (p_T^{\text{reco}}/p_T^{\text{true}}) e^\mu$: **JES closure** = median(r_{corr}) (target ± 1 –2%); **JER** = $(q_{75} - q_{25})/1.349 \div \text{median}$ (Gaussian-equivalent robust width); **pull** = $(y - \mu)/\sigma$ (unit-normal if σ is calibrated). Baselines: B0 = the tuple’s calibrated jets after our cuts; B1 = one global median correction; B2 = the jet-features-only model.

5 Results

5.1 Tier 0 — the uncorrected picture

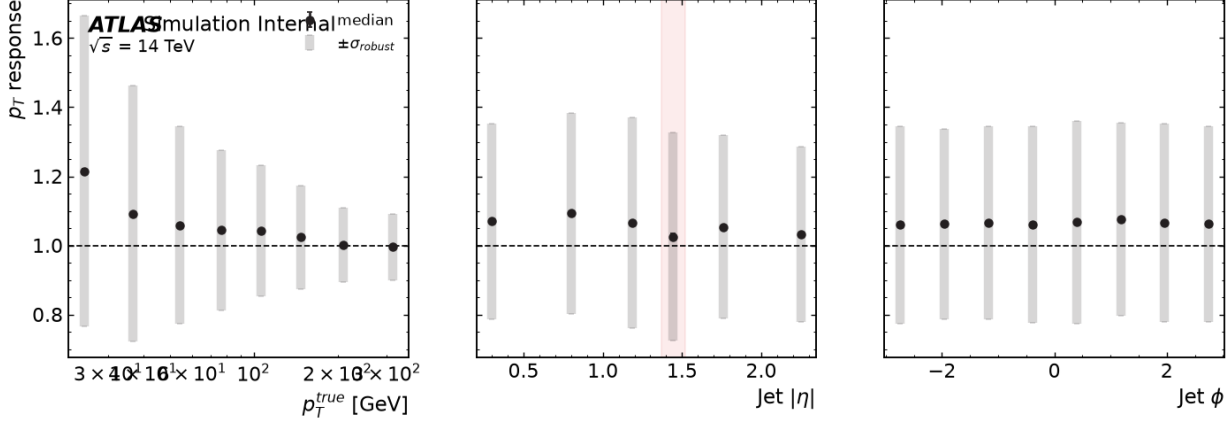


Figure 1: Uncorrected (B0) response versus p_T^{true} , $|\eta|$ (crack band shaded), and ϕ : median with statistical errors and the robust $\pm\sigma$ band. The input jets read +21% high at 20–30 GeV, falling to unity above 150 GeV; ϕ is flat as required in simulation.

5.2 Tiers 1–2 — single inputs and combinations (4-file dataset)

On the initial 4-file dataset (146k jets): the calorimeter neighbor graph beats a set encoding of the same cells on every metric ($\Delta\text{NLL} \approx 0.11$); constituent inputs (tracks or cells) each cut JER by $\sim 30\%$ while the jet 4-vector alone improves resolution by exactly nothing (it can only re-centre); tracks and cells combine (TC 0.147 versus 0.157/0.158 alone); and a median-targeting (MAE) training variant confirmed a $\sim 2\%$ mean-versus-median estimator offset in the NLL closure, fixable by protocol. μ -conditioning changed nothing, isolating the jet-feature advantage as thinned-away constituent information rather than pileup conditioning.

5.3 Tier 3 — the full-data matrix

Scaling to all 17 files improved every configuration and **reversed one Tier-2 conclusion**: with sufficient statistics, cells contribute on top of tracks + jet features (Table 1, Figs. 5–6). The best model, TJC-graph, reaches JER 0.115 at 65–90 GeV versus 0.230 uncorrected.

configuration	NLL	MAE	closure RMS	JER@65–90
TJC-graph	−1.422	0.124	4.0%	0.115
TJC-set	−1.416	0.125	4.2%	0.116
TJ	−1.364	0.130	4.5%	0.125
TC-graph	−1.241	0.153	4.4%	0.130
T	−1.058	0.175	3.0%	0.150
C-graph	−1.086	0.174	6.9%	0.158
B0 uncorrected	—	—	—	0.230

Table 1: Full-data results, isolated test jets, mean of 3 seeds.

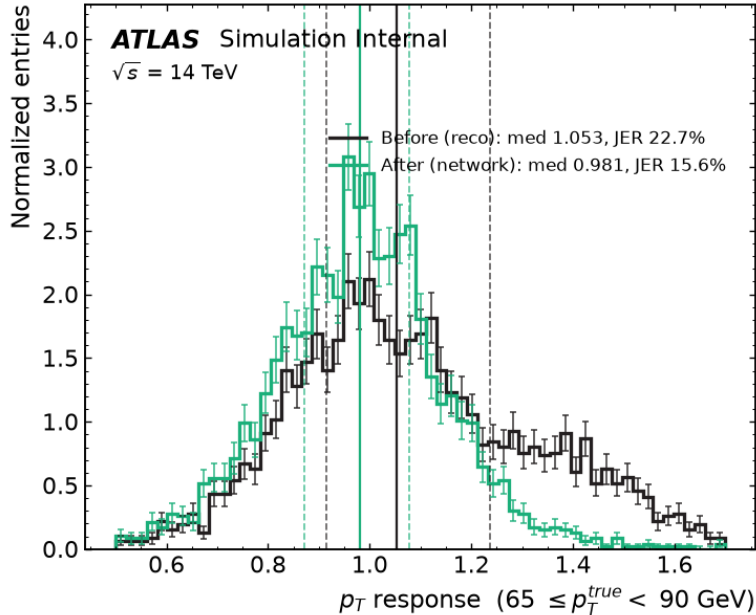


Figure 2: The metric definitions at work in a single bin ($65\text{--}90\text{ GeV}$): response before and after correction, with medians (solid) and quartiles (dashed). The correction both centres the median (JES) and narrows the quartile span (JER).

5.4 Tier 4 — self-supervised pretraining

Masked-cell pretraining (cluster masking at ratio 0.15, reconstructing the scaled cell significance, with a mask-indicator channel making pretrained and finetuned encoders weight-compatible) was performed on all three processed calorimeter corpora ($t\bar{t}$ + single-pion + $hh \rightarrow b\bar{b}\tau\tau$, 14.4k events); scaling the corpus $16\times$ improved validation reconstruction from 0.507 to 0.347. Transfer results (Table 2, Figs. 7–8): finetuning from the pretrained encoder neither helps nor hurts at full labels; the **linear probe**—frozen encoder, trained head—matches full end-to-end training (JER 0.157 vs 0.158) with the best closure of its family; and pretrained initialization is consistently $\sim 1\text{--}2\%$ relatively better at 1% and 10% label fractions.

C-graph variant	NLL	closure RMS	JER@65–90
from scratch	−1.086	6.9%	0.158
pretrained + finetune	−1.086	6.2%	0.158
linear probe (frozen)	−1.041	5.7%	0.157

Table 2: Pretraining transfer, cells-only model.

6 Systematic considerations and known limitations

- **Cell thinning:** the 5D subset carries 75% of jet cell energy with a fluctuating jet-by-jet deficit; cells-only numbers are lower bounds.
- **Truth-match purity:** rests on the recomputed $\Delta R < 0.3$ and $p_T^{\text{true}} > 20\text{ GeV}$ cuts, validated by the B0 response tightening to median 1.065.

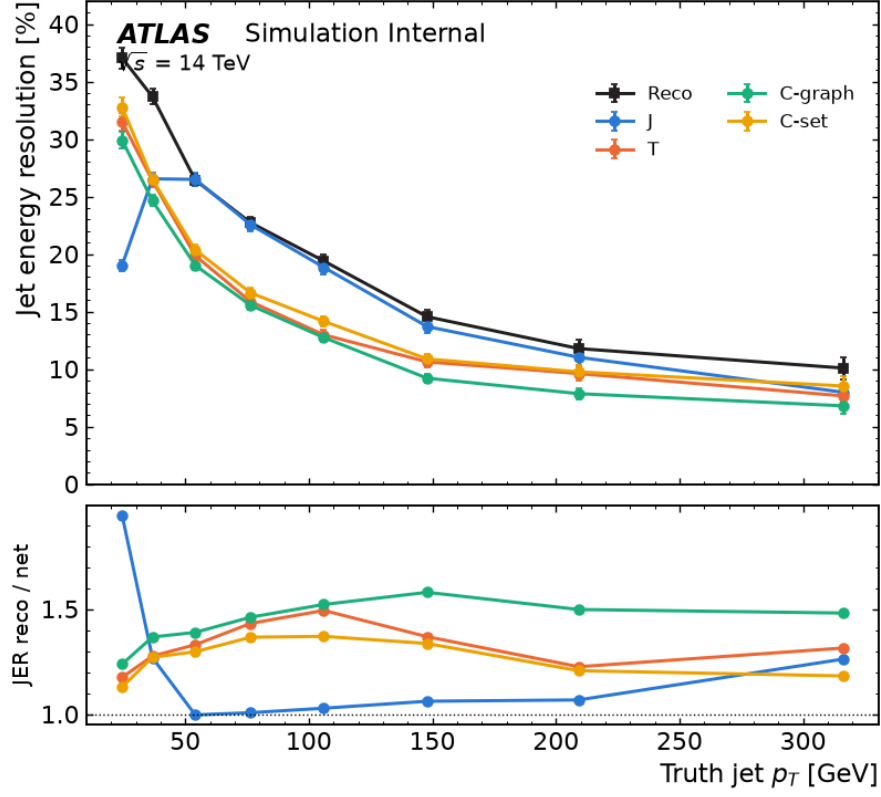


Figure 3: Tier-1 jet energy resolution versus p_T for all single-input configurations with the improvement ratio (uncorrected/model) below. Note the first bin of the jet-features-only model: narrow but at the wrong scale (closure 1.19)—regression to the spectrum prior, and the reason closure and resolution must always be read together.

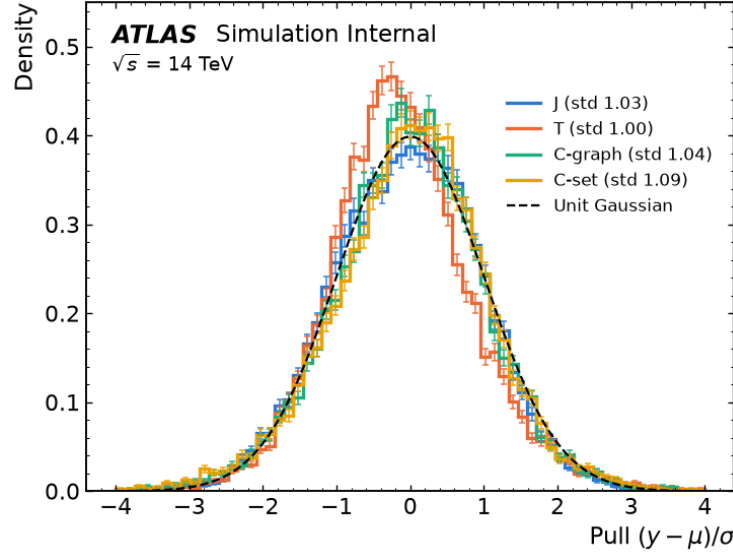


Figure 4: Per-jet uncertainty validation: pull distributions against a unit Gaussian for the Tier-1 models (std 1.03–1.05).

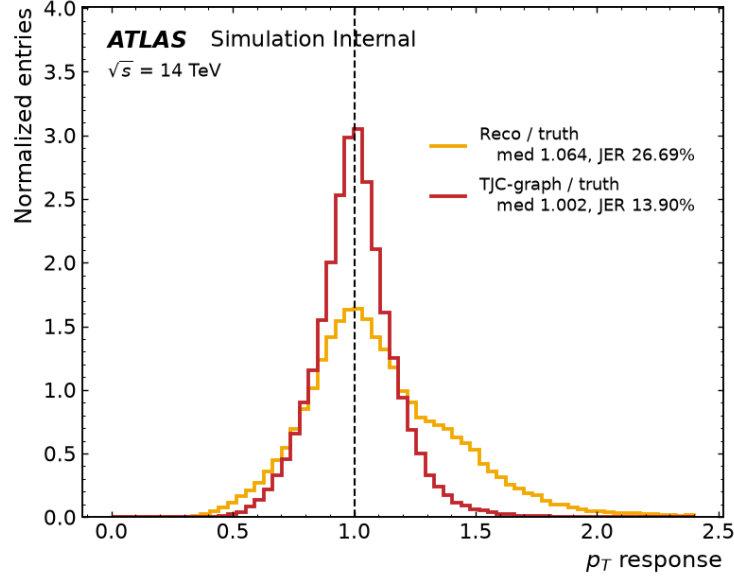


Figure 5: Headline result: inclusive p_T response of the best model (TJC-graph) versus the input reconstructed jets—resolution approximately halved (11.9% vs 23.8%) with median response 1.00.

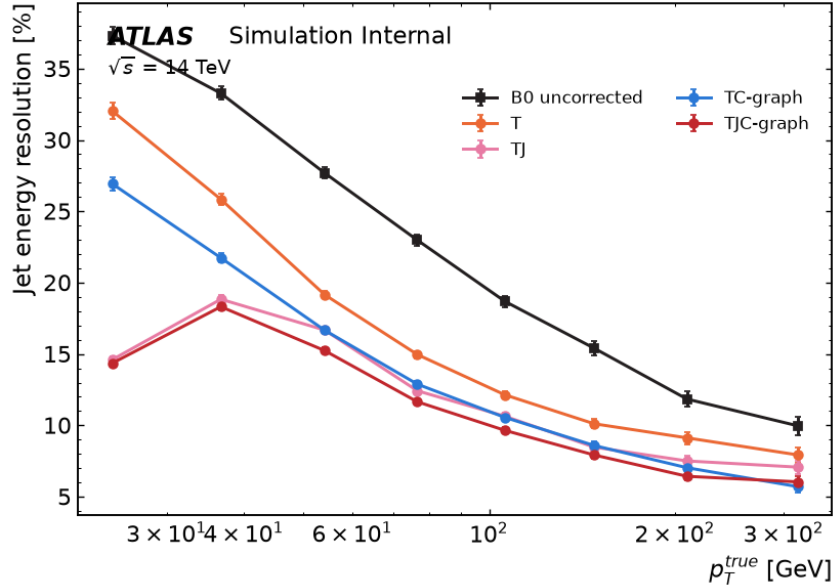


Figure 6: Full-data JER versus p_T for the key configurations: every addition of detector information helps, across the full spectrum.

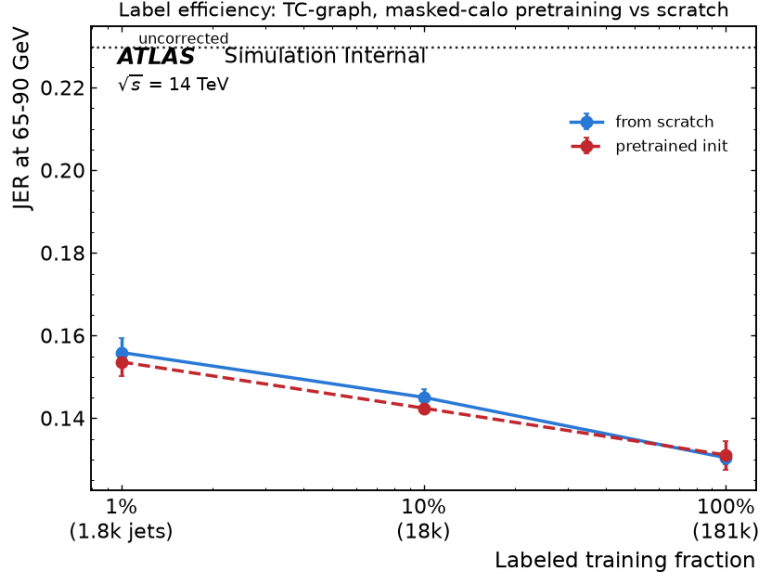


Figure 7: Label efficiency: JER at 65–90 GeV versus labeled training fraction for TC-graph from scratch (blue) and from pretrained initialization (red dashed).

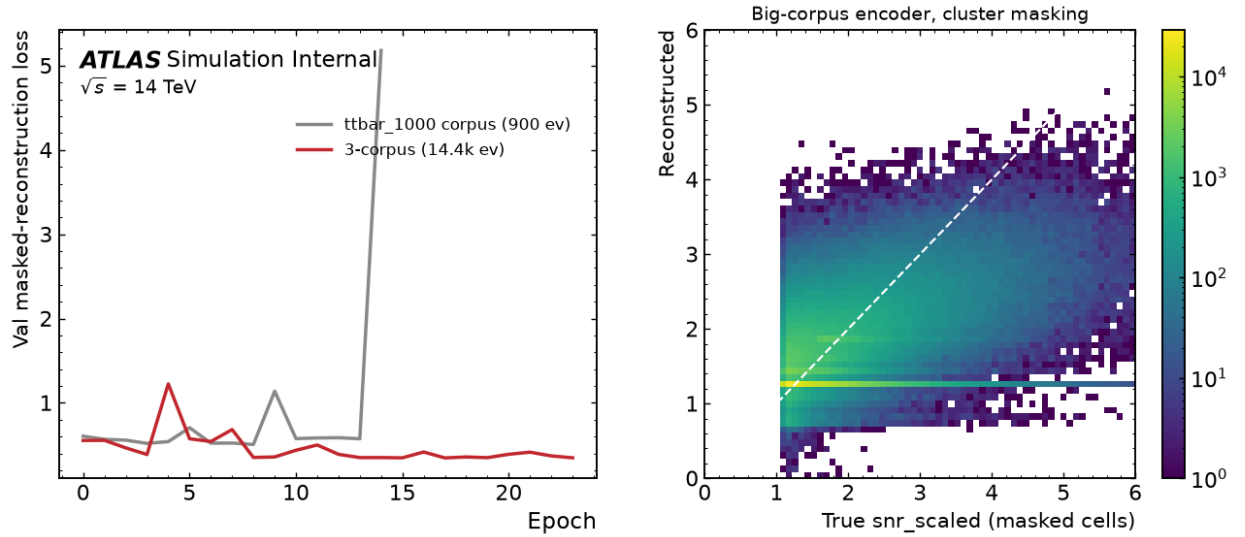


Figure 8: Pretraining QA: validation masked-reconstruction loss for the small and 3-corpus pretrainings (left); reconstructed versus true masked-cell significance for the big-corpus encoder (right).

- **Low- p_T closure:** no configuration closes the 20–30 GeV bin (spectrum-edge migration); numerical inversion is the designated fix, together with the MAE/median protocol for the $\sim 2\%$ NLL estimator offset.
- **Sample dependence:** single $t\bar{t}$ sample; event-level context learned by the models is legitimate physics but sample-specific.
- **Pretraining overlap:** event disjointness between the $t\bar{t}$ pretraining corpus and the 5D files is unverifiable; the 3-corpus mix dilutes, and the probe result is immune in interpretation.
- **Isolation:** headline numbers are for isolated jets; the measured isolated $\rightarrow$ non-isolated degradation defines the target for event-level Stage 2.

7 Conclusions

A shared token encoder over thinned calorimeter cells, tracks, and jet features—with the detector neighbor graph recovered by exact cell matching—halves the jet p_T resolution of the input reconstruction across 20–400 GeV while closing the scale to a few percent. Detector topology is a real inductive bias for calorimeter-only models; full input combination pays off once $\mathcal{O}(10^5)$ labeled jets are available. The masked-cell pretext demonstrably learns calibration-relevant representations (probe $\approx$ full training), with small consistent label-efficiency gains at this model scale. These results establish the baseline and machinery for the flavor-aware studies of Ref. [1] and the joint tagging–calibration study of Ref. [2].

A Operations

The campaign (41 trainings + 2 pretrainings, $4 \times A30$) survived one node freeze requiring an administrator reboot. Root cause: uncapped torch/numpy OpenMP thread pools (56 threads per process, stacked across concurrent jobs starving the scheduler); mitigations now standard: thread caps, niced processes, an alert/pause/kill resource watchdog, SSH connection multiplexing, reboot-proof code and state locations, and re-entrant job queues.

References

- [1] DFM-NOTE-2026-02, *Flavor-Dependent Machine-Learned Jet Calibration*, [notes/DFM-NOTE-2026-02-flavor-calibration/](#).
- [2] DFM-NOTE-2026-03, *Panoptic Flavor Tagging and Jet Calibration*, [notes/DFM-NOTE-2026-03-panoptic-tagging-calibration/](#).
- [3] A. Farbin, *CaloGraphNet reviews*, [reviews/calographnet_review.md](#), [reviews/btagging_masking_review.md](#) (2026).
- [4] ATLAS Collaboration, *Topological cell clustering in the ATLAS calorimeters*, arXiv:1603.02934, and the JES chain references therein.